beyond classical PDEs
conservation laws
shocks, entropy solutions
Hamilton–Jacobi
viscosity solutions
nonlocal & fractional
operators
integrable systems
& solitons
optimal transport
& gradient flows
memory & viscoelastic
transport
microcontinuum continua
nonlinear diffusion
porous medium
delay & retarded
transport
Cattaneo & telegraph
finite-speed diffusion
variational inequalities
hysteresis
variational principles
& Noether
homogenisation
& upscaling
stochastic PDEs
random media
free boundaries
level sets, phase field
MHD & relativistic
transport
numerics & inversion
structure-preserving
discretisations
inverse problems &
data assimilation
machine learning &
operator learning
nonlinear solvers
continuation
linear solvers
preconditioning
adaptivity, moving meshes
parallel decomposition
particle & Monte Carlo
hybrid methods
CFL, stiffness
time stepping
operator splitting
& multiphysics coupling
discretisation: consistency
stability, convergence
verification & validation
numerical diffusion
& dispersion
limiters · TVD · WENO
bridges to other fields
finance
images & signals
networks & traffic
machine learning & data
one mathematics of transport
quantum & condensed matter
geoscience & climate
biology & medicine
plasma & astrophysics
engineering devices
foundations
PDE taxonomy
+ boundary conditions
higher order
& dispersive
entropy & variational
structure
dimensions, geometry
& manifolds
Fourier, Green functions
propagators
material derivative
Lagrangian ↔ Eulerian
memory & linear response
Mori–Zwanzig
dimensionless numbers
Pe · Re · Ra · Da · Kn
the balance law
flux · divergence
constitutive laws
Fick · Fourier · Darcy
random walks, Langevin
fluctuation–dissipation
bulk fluid transport
non-Newtonian rheology
vorticity & pressure
shear dispersion
Taylor–Aris
Darcy & porous flow
convection & buoyancy
Rayleigh–Bénard, Marangoni
turbulence &
eddy diffusivity
jets · plumes · wakes
boundary layers
advection
mixing & chaotic
advection
recurring mechanisms — the shapes that keep returning
nonlinearity &
mode coupling
scaling & self-
similarity
duality &
transforms
equilibria &
critical points
linearisation
& stability
bifurcation
theory
chaos &
predictability
phase transitions
& criticality
the intersections
map
noise-induced
phenomena
bibliography
(primary · review · sceptical)
kinetic & radiative
radiative transfer
neutron transport
Boltzmann equation
H-theorem
lattice-Boltzmann
particle methods
Fokker–Planck
Kramers escape
BGK · Vlasov
plasmas
interaction / source / sink
accretion · ablation
surface growth
autocatalysis &
oscillations
reaction–diffusion
Turing patterns
chemotaxis
Keller–Segel
wells · injection · efflux
breakthrough
subsidence &
poroelasticity
phase change
Stefan problem
fronts & travelling
waves
absorption · dissolution
interfacial transfer
reaction kinetics
Arrhenius
viscous dissipation
Joule heating
sedimentation · deposition
filtration
sorption & retardation
microscopic transport
conduction (Fourier)
diffusion coefficients
from kinetic theory
Knudsen & rarefied
transport
diffusion (Fick)
Nernst–Planck &
electrochemical
Soret · Dufour
Maxwell–Stefan
anomalous diffusion
CTRW, fractional
Brownian motion
Einstein relations
universality — the strongest claim
the atlas (master note)
physics, chemistry
& field theory
mathematics itself
(geometric flows, categories)
why everything flows
(five universal structures)
the translation lookup
(object → transport form)
computation & algorithms
(Dijkstra, PageRank, SGD)
limits of the claim
(where it stops)
information, statistics
& inference
cognitive science & mind
(decisions, fields, memory)
social behaviour, economics
& language
panta rhei — all things flow
collective behaviour
& synchrony
crossings & cross-
fertilisation
the river
as a model
beauty & the flow
(marbling, patterns)
Heraclitus
& the river
irreversibility &
the arrow of time